

UBT518: BIOSTATISTICS AND OMICS LAB

L	T	P	Cr
0	0	3	1.5

Course objective: The course aims to make students understand and apply biostatistics and bioinformatics tools for storage, retrieval and analysis of biological sequence and structure data. The course will allow students to learn application of statistics on biological data.

Detail contents:

Biostatistics

Descriptive statistical analysis of given data (Measures of central tendency and measures of dispersion and variability), Analyse relationship between tow variables by correlation and regression analysis, Statistical tests (t-test, ANOVA, Chi-square test)

Omics

To explore DNA sequence databases such as GenBank, EMBL, NCBI; Retrieve, FASTA sequence from GenBank database and perform sequence analysis, To explore protein sequence databases such as UniProt, PROSITE, PFAM and retrieve comprehensive information of give protein, To study protein 3D structure databases using PDB, PDBsum, CATH and SCOP, Database, To study pairwise, global and multiple sequence alignment: Emboss Needle, Emboss Water, Clustal Omega, MUSCLE, MAFFT and T-COFFEE, To perform homology searches using BLAST and its variants, Construction of phylogenetic tree using MEGA software, To find interactions among proteins using STRING, Retrieving DNA from UCSC genome browser and NCBI and Gene prediction using AUGUSTUS, To search CpG islands and ORF in genomic DNA sequences, To search for open reading frames in a given DNA sequence, To predict post translational modifications in given protein sequences, To predict RNA structure, To predict tertiary structure for an amino acid sequence using SWISS-MODEL,

Course Learning Outcomes (CLO):

The students will be able to

1. Explore biological databases and retrieve relevant data from them

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

2. Perform pairwise and multiple sequences alignment, construct phylogenetic tree, analyze and make inferences from it.
3. Analyze sequence and structure of bio-macromolecule data.
4. Analyze biological data using basic statistical concepts such as measure of central tendencies, measure of dispersion and sampling.
5. Perform statistical hypothesis testing using tools such as t-test, ANOVA, Chi-square test

Reference Books:

1. Mount D W, Bioinformatics - Sequence and Genome Analysis, Cold Spring Harbour Laboratory Press (2001)
2. Higgins, D. and Taylor, W., Bioinformatics: Sequence, Structure and Databanks A Practical Approach, Oxford University Press (2000).
3. Zar, J. H., Biostatistical Analysis. Pearson Education (2014) 5th Edition.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	Sessionals (May include assignments / quizzes / performance)	100

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